FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2023

TEST DATE: 22-03-2023

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

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Physics

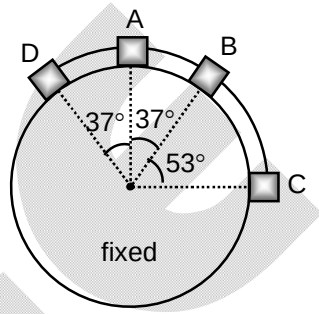
PART – A

SECTION – A

(Single Choice Answer Type)

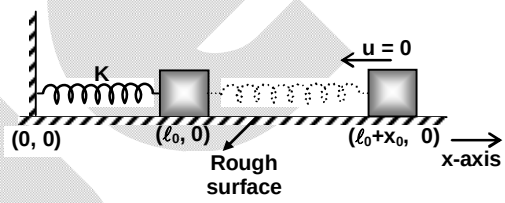
This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. Figure shows four small blocks interconnected with light strings and placed over a fixed sphere as shown. Masses of blocks $M_A = M_B = M_C = 1$ kg and $M_D = 2$kg. The coefficient of friction between blocks and the sphere is $\mu = 0.5$. The system is released from the position shown. The friction force acting between block D and sphere is (ignore dimensions of blocks)



(A) 2 N (B) 7 N
(C) 5 N (D) 8 N

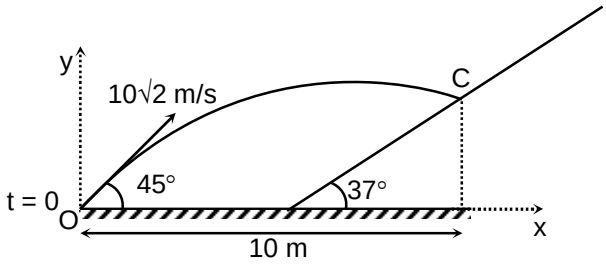
2. A small block connected with a spring of force constant k and natural length ℓ_0 is placed over a rough horizontal surface. The other end of spring is fixed to a rigid wall as shown in the figure. The block is pulled to right to give the spring an elongation x_0 and released. The block comes to rest at position where spring is relaxed. The coordinate where kinetic energy of block is maximum and maximum value of kinetic energy are respectively.



(A) $\left(\ell_0 + \frac{x_0}{4}, 0\right)$ and $\frac{1}{4}kx_0^2$ (B) $\left(\ell_0 + \frac{x_0}{3}, 0\right)$ and $\frac{1}{3}kx_0^2$

(C) $\left(\ell_0 + \frac{x_0}{2}, 0\right)$ and $\frac{1}{4}kx_0^2$ (D) $\left(\ell_0 + \frac{x_0}{2}, 0\right)$ and $\frac{1}{8}kx_0^2$

3. At time t = 0, a particle is projected from point O on the ground with velocity $10\sqrt{2}$ m/s at an angle 45° with horizontal. It strikes a long fixed smooth inclined plane of inclination 37° at point C which is horizontally 10 m away from point O as shown in the figure. Assuming the collision to be perfectly inelastic, the time at which particle reaches the maximum height from the ground is ($g = 10$ m/s^2)

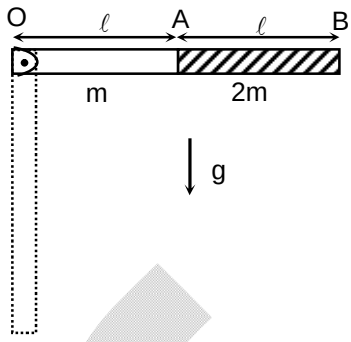


(A) $\frac{7}{3}$s (B) $\frac{10}{3}$s

(C) $\frac{16}{3}$s (D) $\frac{20}{3}$s

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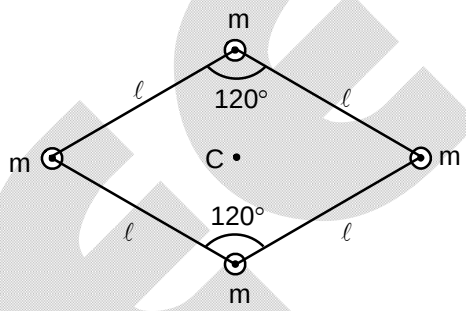
4. A composite rod made up of two uniform segments of mass m and 2m each of length ℓ is hinged at end O. The rod is released from horizontal position in vertical plane. The force experienced by segment AB due to remaining rod, when rod becomes vertical is



(A) $\frac{11}{2}$mg (B) $\frac{7}{4}$mg

(C) $\frac{5}{2}$mg (D) $\frac{31}{5}$mg

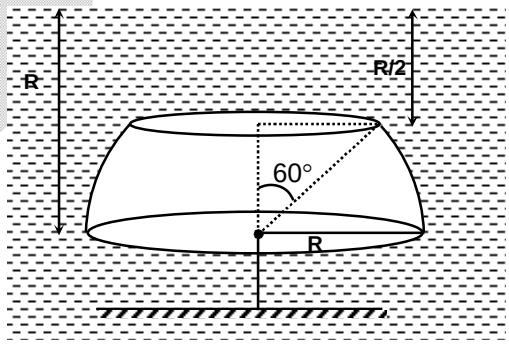
5. Consider four point masses each of mass m are fixed at the vertex of a Rhombus of side ℓ. The escape speed from the centre of the Rhombus is (G is universal gravitational constant)



(A) $\sqrt{\frac{8Gm}{\ell}}$ (B) $\sqrt{4\sqrt{2}\frac{Gm}{\ell}}$

(C) $\sqrt{\frac{8Gm}{\ell}\left(1+\frac{1}{\sqrt{3}}\right)}$ (D) $\sqrt{\frac{8Gm}{\ell}\left(1+\frac{1}{\sqrt{5}}\right)}$

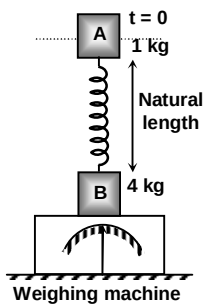
6. A part of solid wooden hemisphere of radius R, supported by a string inside water as shown in the figure. In equilibrium, the base of the hemisphere is at a depth of R below the water surface. The force applied by the water on curved surface is (Take density of water ρ, density of wood σ and atmospheric pressure P_0)



(A) $P_0\left(\frac{\pi R^2}{4}\right)+\frac{1}{6}\rho g\pi R^3$ (B) $P_0\left(\frac{\pi R^2}{2}\right)+\frac{1}{3}\rho g\pi R^3$

(C) $P_0\left(\frac{\pi R^2}{4}\right)-\frac{1}{6}\rho g\pi R^3$ (D) $P_0\left(\frac{\pi R^2}{2}\right)+\frac{1}{2}\rho g\pi R^3$

7. Two blocks having mass M_A = 1 kg and M_B = 4 kg are connected by spring and placed vertically on a weighing machine. Block A is held so that the spring is relaxed. At time t = 0 block A is released from this position and performs simple harmonic motion with angular frequency ω = 25 rad/s. The reading of weighing machine at time t is (g = 10 m/s^2)

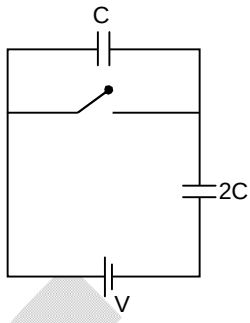


(A) 50 + 10cos(25t) (B) 50 – 10cos(25t)

(C) 40 – 10sin(25t) (D) 60 + 10cos(25t)

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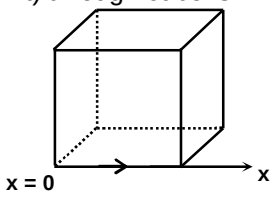
8. Figure shows a circuit consists of two capacitors C and 2C and ideal battery of terminal voltage V. Initially the switch was open for a long time and capacitors were fully charged. The heat developed in the circuit after switch is closed.

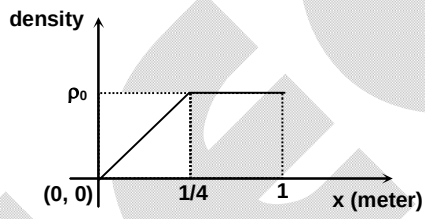


(A) $\frac{1}{4}CV^2$ (B) $\frac{1}{3}CV^2$

(C) $\frac{3}{4}CV^2$ (D) $\frac{2}{3}CV^2$

9. The volume charge density as a function of distance x from one face inside a cube of side 1 meter is varying as shown in the figure. For $\rho_0 = 8.85\times10^{-12}$ C/m^3. The total electric flux (in S.I. unit) through cube is

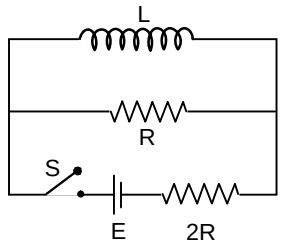




(A) $\frac{1}{2}$ (B) $\frac{3}{4}$

(C) $\frac{7}{8}$ (D) $\frac{7}{4}$

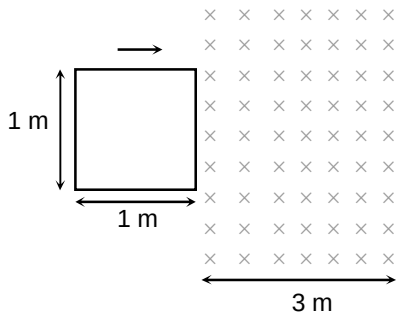
10. An inductor of inductance L and resistors of resistances R and 2R are connected to a battery of emf E. The internal resistance of battery is negligible. The switch S is closed at t = 0. The current through resistor R as a function of time is:



(A) $\frac{E}{2R}e^{-\frac{2Rt}{3L}}$ (B) $\frac{E}{3R}e^{-\frac{2Rt}{3L}}$

(C) $\frac{E}{2R}\left[1-e^{-\frac{2Rt}{3L}}\right]$ (D) $\frac{E}{3R}\left[1-e^{-\frac{2Rt}{3L}}\right]$

11. A square loop of side 1 m being moved towards right at a constant speed of 1 m/s. At t = 0, the front edge of loop enters the 3 m wide uniform magnetic field B = 2 Tesla. The resistance of square loop is 2Ω. The total heat produced in the loop during the interval 0 to 5 sec is

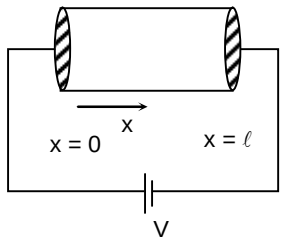


(A) 4 Joule (B) 2 Joule

(C) 1 Joule (D) 8 Joule

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12. Figure shows a cylindrical conductor of length ℓ and area of cross-section A. Its conductivity changes with distance (x) from one of its end as $\sigma = \sigma_0 \frac{\ell}{x}$ (σ_0 is a constant). A cell of emf V is connected across its ends. The electric field as a function of x is



(A) $\frac{Vx}{2\ell^2}$ (B) $\frac{Vx}{\ell^2}$

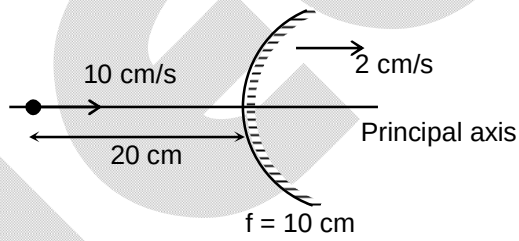
(C) $\frac{2Vx}{\ell^2}$ (D) $\frac{4Vx}{\ell^2}$

13. In amplitude modulation, the carrier signal is given by C(t) = 10 sin(20000πt) and modulating signal is given by m(t) = 5sin(2000πt). The values of modulation index and Bandwidth of the modulated signal are respectively.

(A) 0.1 and 200 Hz (B) 0.5 and 2000 Hz

(C) 0.2 and 1000 Hz (D) 0.4 and 2000 Hz

14. A convex mirror of focal length f = 10 cm moving with velocity 2 cm/s along principal axis as shown in the figure. For the given instant, an object moving on principal axis with velocity 10 cm/s and at a distance 20 cm from pole of mirror. The velocity of image at this instant is



(A) $\frac{9}{10}$ cm/s (B) $\frac{8}{9}$ cm/s

(C) $\frac{10}{9}$ cm/s (D) $\frac{9}{8}$ cm/s

15. Work function of metal A is equal to ionization energy of Hydrogen atom in its ground state. Work function of metal B is 7.2 eV. Light of same wavelength is incident on A and B. The maximum kinetic energy of emitted photoelectrons from B is twice that of photoelectrons emitted from A. The wavelength (in Å) of incident light is (Take hc = 12400 eVÅ)

(A) 615 (B) 618

(C) 620 (D) 624

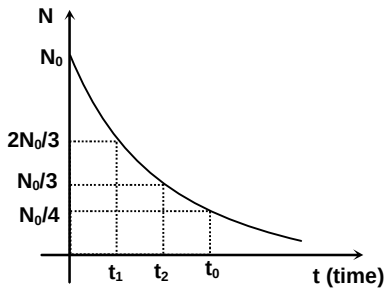
16. For a radioactive element, number of nuclei plotted with time as shown in the figure. If t_0 = 40 minute, the value of $t_2 - t_1$ is

(A) 10 minutes

(B) 20 minutes

(C) 30 minutes

(D) 5 minutes



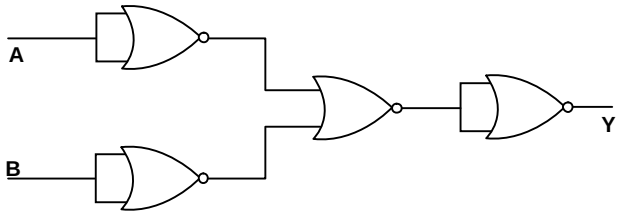
17. If the constant of gravitational constant (G) and Plank's constant (h) and the velocity of light (c) be chosen as fundamental units. The dimensions of the radius of gyration is:

(A) $h^{1/2}c^{-3/2}G^{1/2}$ (B) $h^{1/2}c^{3/2}G^{1/2}$

(C) $h^{1/2}c^{-3/2}G^{-1/2}$ (D) $h^{-1/2}c^{-3/2}G^{1/2}$

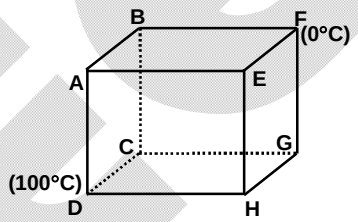
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18. Identify the logic operation carried out by the given circuit.
(A) XOR Gate
(B) NAND Gate
(C) AND Gate
(D) NOR Gate



19. A vernier calliper with no zero error is used to measure the diameter of a cylinder. It is found that the zero of the vernier scale lies between 6.10 and 6.15 cm of the main scale. The vernier scale has 50 divisions equivalent to 2.35 cm. The 24th division of the vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is
(A) 6.137 cm (B) 6.172 cm
(C) 6.174 cm (D) 6.147 cm

20. Figure shows twelve identical rods made up of same material form a cube. The junctions D and F are maintained at 100°C and 0°C respectively. In steady state, the temperature of the junction H is
(A) 30°C (B) 60°C
(C) 40°C (D) 50°C

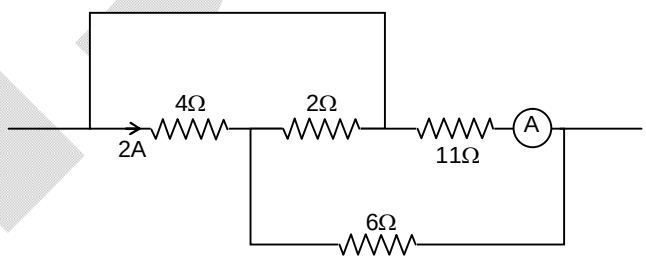


SECTION – B

(Numerical Answer Type)

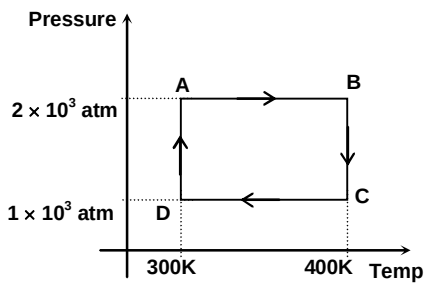
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A part of large electrical circuit is shown in the figure. The current passing through 4Ω resistor is 2A. Find the reading of ideal ammeter (in ampere)?



22. A well insulated calorimeter contains a mixture of water and ice. A heater supplies energy at the rate of 30 watt to the mixture. At the ends of second, third and fourth minute after heater is switched on, temperature of the mixture is 0°C, 2°C and 8°C respectively. How many grams of ice were initially present in the calorimeter? (Latent heat of melting of ice = 400 J/g and Specific heat of water = 4J/g°C)

23. Figure shown the cyclic process (ABCDA) for $\frac{2}{R}$ moles of an ideal gas. Calculate the net work done by the gas in joule for complete cycle. (Take $\ell n2 = 0.7$, R = universal gas constant)

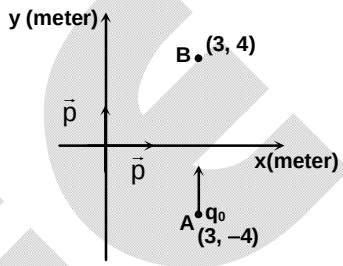


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24. A compound microscope consists of an objective lens of focal length 5 cm and an eyepiece of focal length 6.25 cm separated by a distance of 15 cm. At what distance (in cm) from objective lens should an object be placed in order to obtain the final image at least distance of distinct vision (25 cm)?

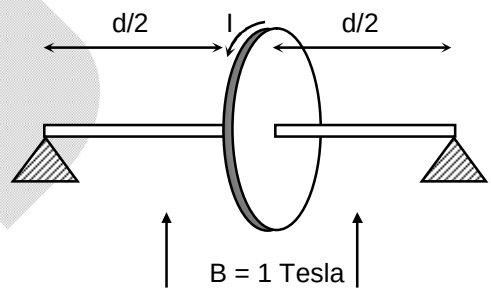
25. Unpolarized light passes through a system of 5 polaroids, which are arranged in succession such that axis of each polaroid is turned through 30° with respect to the preceding one. What percentage of incident intensity will come out through the system? (Rounded off to the nearest integer)

26. Two short electric dipoles each of dipole moment P are placed at origin with direction along the coordinate axis as shown in the figure. Calculate the work done (in joule) in moving a point charge q_0 from coordinate (3, –4) to (3, 4)? (Take $\frac{Pq_0}{4\pi\varepsilon_0} = 125$ SI unit)

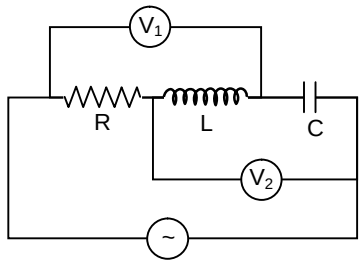


27. A radiation is emitted by 100 watt bulb and it generates an electric field and magnetic field. 2.4% of the power is converted into radiation. Calculate the peak value of electric field (V/m) at a distance 2m from the bulb? (Take, $\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ Nm^2/C^2 and speed of light = 3×10^8 m/s)

28. A wooden disc of mass M = π kg and radius R = 50 cm is mounted on a massless rod of length d = 40 cm. The disc has a single loop of current carrying wire wound on its circumference. The ends of the rod are supported at its end so that rod is horizontal and disc is vertical. A uniform magnetic field B = 1 Tesla exist in vertically upward direction. Find the least value of current (ampere) in the wire for which one end of the rod leaves the support? (Take g = 10 m/s^2)

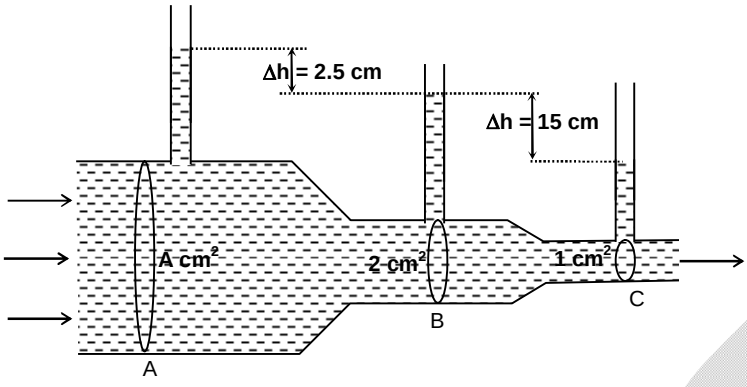


29. In the series LCR circuit, the voltmeters reading are V_1 = 100 V and V_2 = 120 V as shown in the figure. The voltage of AC source is 130 V. The power factor of the circuit is $\frac{x}{13}$. Calculate the value of x?



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30. Water is flowing in a tube of varying cross-section. Difference in water levels are shown in the vertical tubes. The area of cross section A, B and C are A cm^2, 2 cm^2 and 1 cm^2 respectively. The speed of water at cross-section A is $\frac{1}{\sqrt{x}}$ m/s. Find the value of x. ($g = 10\ m/s^2$)



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Chemistry

PART – B

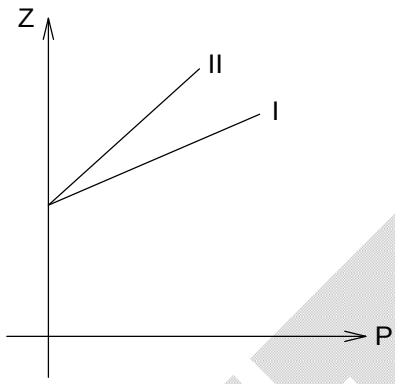
SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

31. In the following molecular/ions
$O_2^+, O_2^{-2}, C_2^{-2}, N_2^{-2}, N_2^{-4}$
Which one is paramagnetic and has highest bond length?
(A) O_2^+ (B) C_2^{-2}
(C) O_2^- (D) N_2^{-4}

32. The equation of state for 1 mole of Ne and 1 mole of Xe is given as
$P(V- b) = RT$, at temperature T. (b = vander Waals constant)
Compressibility factor Z vs P curve is given as



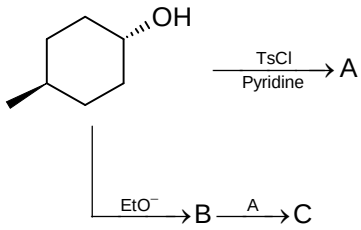
(A) Curve I – Ne (B) Curve II – Xe
(C) Curve I – Xe (D) cannot be predicted

33. Which of the following is exothermic in nature?
(A) $Na(g) + e \longrightarrow Na^-(g)$ (B) $Ne(g) + e \longrightarrow Ne^-(g)$
(C) $N(g) + e \longrightarrow N^-(g)$ (D) $O^-(g) + e \longrightarrow O^{2-}(g)$

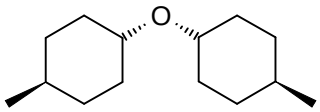
34. Ozone is not used as
(A) Germicide (B) Sterilising water
(C) Bleaching oils (D) Solvent

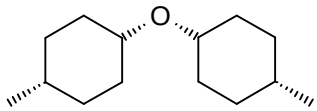
35. Difference between the radii of 2nd and 3rd orbits of He^+ is ΔR_1. The difference between the radii of 3rd and 4th orbits of Li^{+2} is ΔR_2. Find $\Delta R_1 / \Delta R_2$
(A) $\frac{21}{10}$ (B) $\frac{10}{21}$
(C) $\frac{14}{15}$ (D) $\frac{15}{14}$

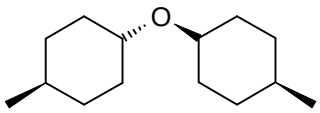
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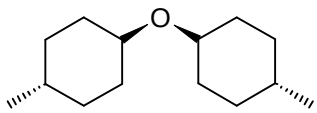
36. 

Find product (C) will be

(A) 

(B) 

(C) 

(D) 

37. Identify correct statements:
(A) Biuret test detects presence of glucocide linkages.
(B) Barfoed's test based on reduction of ferric acetate to ferrous oxide.
(C) Molisch's test is given by amino acids.
(D) Molisch's test is based on dehydration of carbohydrate by sulphuric acid to produce an aldehyde.

38. Polymer which is used in manufacture of paints and lacqers?
(A) P.V.C. (B) Polypropene
(C) Bakelite (D) Glyptal

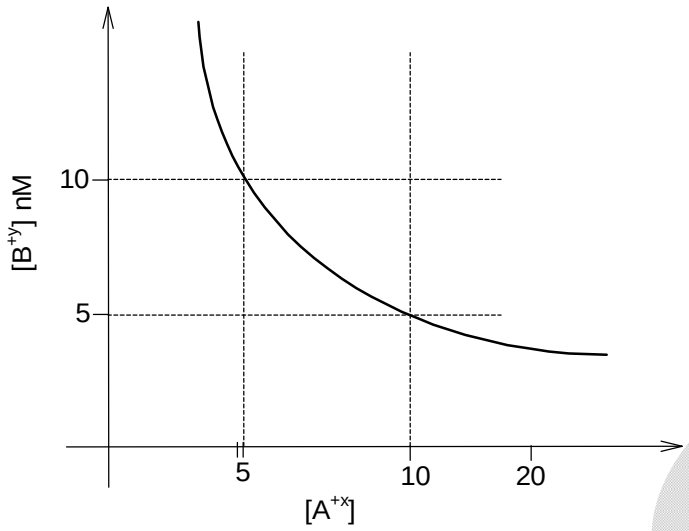
39. Increased fragility of RBC's and muscular weakness is due to deficiency of Vitamins?
(A) Vitamin B_{12} (B) Vitamin D
(C) Vitamin E (D) Vitamin K

40. Antioxidants are important and necessary food additives. Find anti oxidant from the following
(A) Sodium benzoate (B) Table salt
(C) Butylated hydroxyl toluene (D) Aspartame

41. Example of hexagonal crystal system is
(A) CdS (B) H_3BO_3
(C) ZnS (D) $BaSO_4$

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42. Find out the formula and solubility product constant of salt if concentration of ions (solubility of A ion and B ion) curve is given below (concentration is in nano molar)



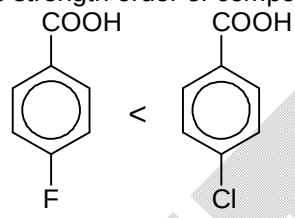
(A) AB_2, 5×10^{-18} M^3

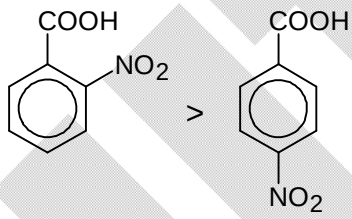
(B) A_2B, 2.5×10^{-28} M^3

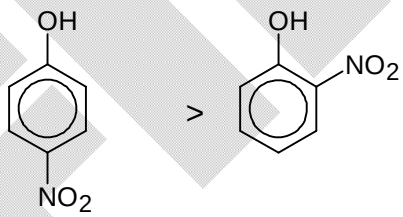
(C) AB, 5×10^{-17} M^2

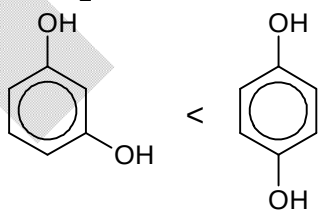
(D) A_2B_3, 2.5×10^{-35} M^5

43. Acidic strength order of compounds is given below. Identify incorrect option.

(A) 

(B) 

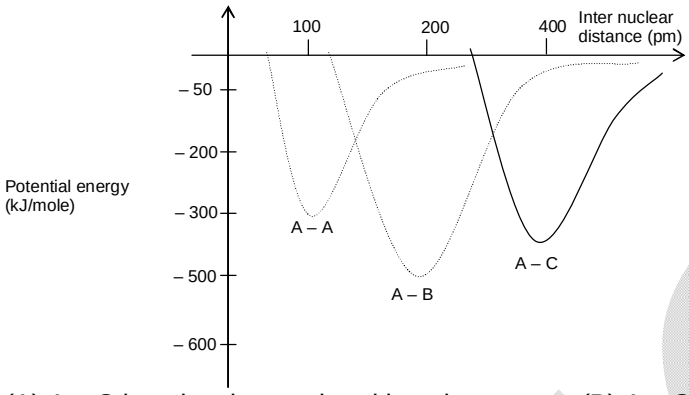
(C) 

(D) 

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44. Identify incorrect statements
(A) In froth floatation process cresols and aniline are used as froth stabilizers.
(B) Calcination removes the volatile matter like CO_2, $H_2O(g)$.
(C) Reverberatory furnace is used in purification of AuCl.
(D) Mond's process is used for refining of nickel.

45. Intermolecular potential energy for the molecules A – A, A – B and A – C given below suggest that:



(A) A – C has the shortest bond length.
(B) A – C has lowest bond enthalpy.
(C) A – A has most ionic compound.
(D) A – B has the stiffest bond.

46. Chlorination of organic compound is given as follows

$$CH_3-CD_2-CH_3+Cl_2 \xrightarrow{h\nu} \underbrace{H_3C-\underset{\displaystyle Cl}{\underset{|}{CD}}-CH_3}_{50\%} + DCl + \underbrace{CH_3-CD_2-CH_2-Cl}_{50\%} + HCl$$

What is the reactivity ratio of D : H for above reaction?
(A) 2 : 3
(B) 3 : 2
(C) 1 : 3
(D) 3 ; 1

47. Ozone layer protects us from the Sun's radiation it's wave length lies in the region
(A) 600 – 750 pm
(B) 200 – 315 nm
(C) 800 – 100 nm
(D) 0.3 – 0.5 pm

48. Identify incorrect statement
(A) Photochemical smog has main component ozone, nitric oxide, acrolein and PAN.
(B) H_2O_2 is used for the purpose of bleaching clothes in process of laundary.
(C) Herbicides are sodium chlorate, sodium arsinite, etc.
(D) Aldrin and Dieldrin were replaced by DDT in the market by pesticide industry.

49. For which of the following processes, ΔS is negative
(A) Heating of an iron rod
(B) $C_{(Diamond)} \longrightarrow C_{(Graphite)}$
(C) $O_2(g, 1\ atm) \longrightarrow O_2(g, 2\ atm)$
(D) $N_2(g) \longrightarrow 2N(g)$

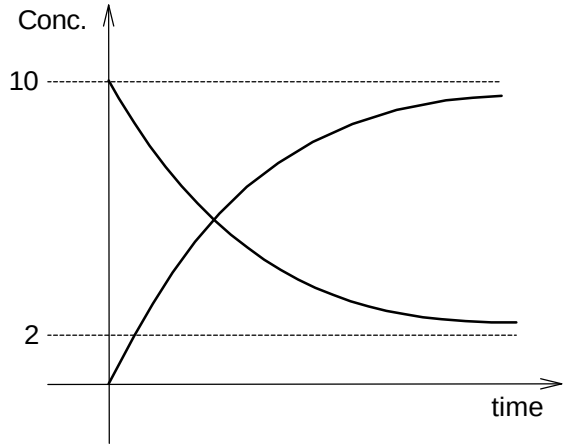
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50. For the chemical reaction at temperature 298 K,

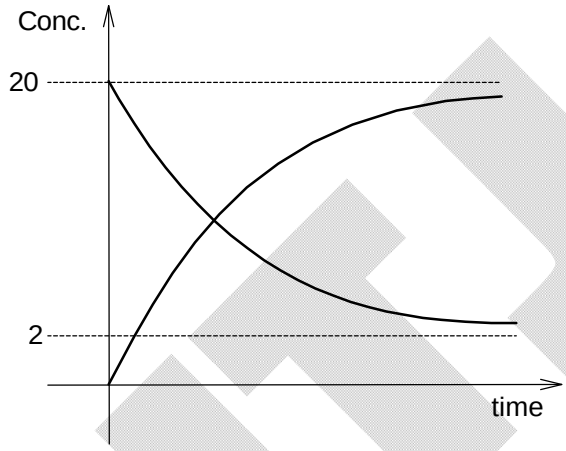
$A(g) \rightleftharpoons B(g) \quad \Delta G^{o} = -413.15 \text{ cal / mole}$

Concentration (mole/lire) vs. time graph is given below. Identify the correct graph for the above reaction.

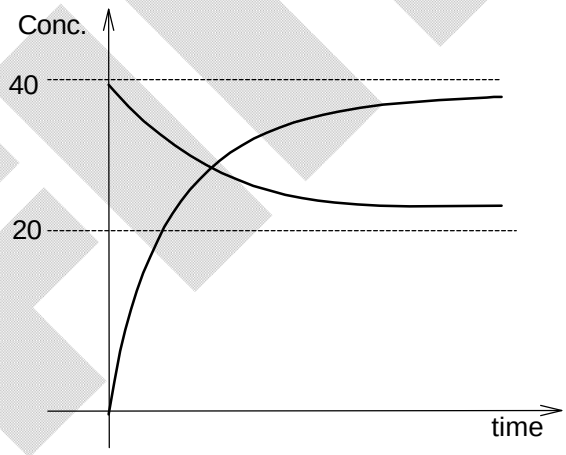
(A)



(B)

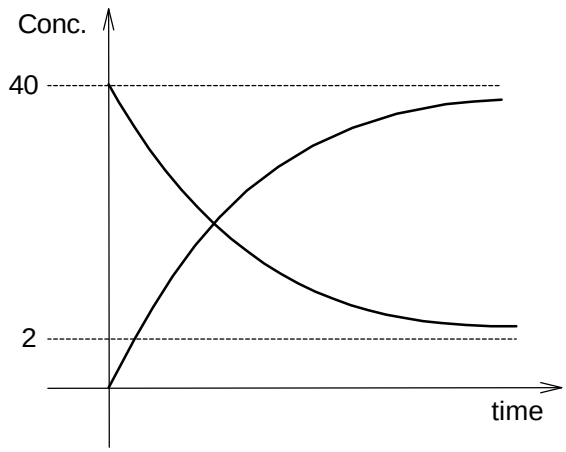


(C)



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(D)

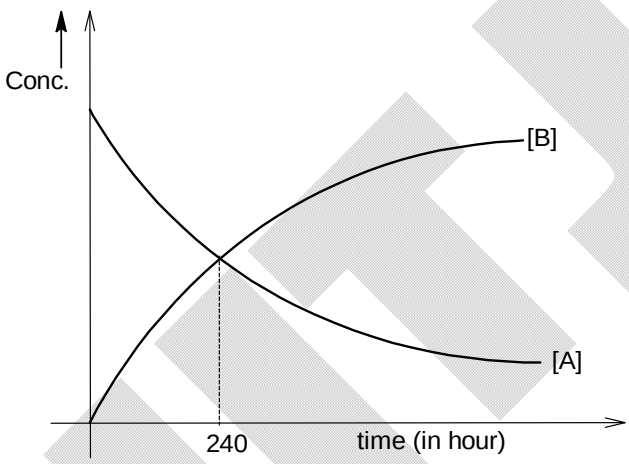


SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. For first order reaction $3A \longrightarrow B$, concentration vs. time curve is given below:



Find half-life of A in days.

52. x and y are the total number of all possible isomer of C_4H_7Cl (including all stereoisomers) in alkene and cyclic form respectively. Find out x – y.

53. 0.5 gm mixture of ethanal and acetone heated with Fehling solution gave 0.975 gm. of red precipitate. If x is the percentage of acetone in the mixture, find $\frac{x}{10}$? (Atomic mass of Cu = 63.5)

54. Total number of position isomers of B_3N_3XY will be (if X = Cl, Y = F)

55. On quantitative reaction of urea with sodium hydroxide, NH_3 is released which is neutralized by 100 ml of 2 M HCl. Find out amount (gram) of urea required.

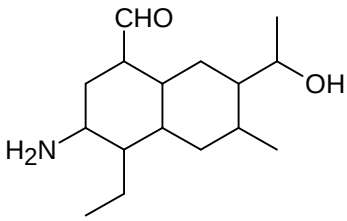
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56. If 'a' gram of non-volatile solute (A) is dissolved in 'b' gram of pure liquid (B), the vapour pressure becomes $\frac{9}{10}$th of it's original value. The molecular weight of the solvent (B) is $\frac{1}{5}$th of the molecular weight of solute (A). Find (b/a)?

57. Enthalpy change (in kcal) for a hypothetical reaction
$AB_2(g) \longrightarrow A^+(g) + B^-(g) + B(g)$ is x kcal/mole
Find out x/100?
(Average bond enthalpy of A – B = 100 Kcal/mole, 1st ionization enthalpy of A(g) = 250 Kcal/mole, Electron gain enthalpy of B(g) = - 100 Kcal/mole)

58. 2 gram of O_2 and some NaCl is formed by heating sample of $NaClO_3$. The residue is reacted with excess of $AgNO_3$ and form AgCl. Mass of AgCl (in gram) will be (answer in nearest integer).

59. Find out the total number of hydrogen atoms attach to chiral carbon atom for given molecules.



60. Total number of nitrogen atoms in the structure of Iproniazid.

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Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

61. A box contain 24 identical balls of which 12 are white and 12 are black. The balls are drawn at random from the box one at a time with replacement. The probability that a white ball is drawn for the 4th time on the 7th draw is

(A) $\frac{5}{64}$ (B) $\frac{27}{32}$

(C) $\frac{5}{32}$ (D) $\frac{1}{2}$

62. Let A and B are two independent events and C is an event in which exactly one of A or B occur, then $\frac{P(C)}{(P(A\cap B))(P(A'\cap B'))}$ =

(A) one (B) less or equal to one

(C) greater or equal to one (D) none of these

63. If the sum of all the solutions of the equation $8\cos x\left(\cos\left(\frac{\pi}{6}+x\right)\cos\left(\frac{\pi}{6}-x\right)-\frac{1}{2}\right)-1=0$ in $[0, \pi]$ is $k\pi$, then k =

(A) $\frac{13}{9}$ (B) $\frac{8}{9}$

(C) $\frac{20}{9}$ (D) $\frac{2}{3}$

64. Let $S = \{t \in R : f(x) = |x - \pi| (e^{|x|} - 1) \sin|x|$ is not differentiable at t}, then set S is equal to

(A) {0} (B) $\{\pi\}$

(C) $\{0, \pi\}$ (D) none of these

65. Let A $(\alpha, -2)$, B$(\alpha, 6)$ and C$\left(\frac{\alpha}{4}, -2\right)$ be vertices of ΔABC. If $\left(5, \frac{\alpha}{4}\right)$ is the circum centre of ΔABC, then which of the following is not correct about ΔABC

(A) area is 24 (B) perimeter is 25

(C) circum radius is 5 (D) inradius is 2

66. Let $I_n = \int \tan^n x\, dx$ $(n > 1)$ and $I_4 + I_6 = a\tan^5 x + bx^5 + c$ where c is a constant of integration, then the ordered pair (a, b) is equal to

(A) $\left(-\frac{1}{5}, 0\right)$ (B) $\left(-\frac{1}{5}, 1\right)$

(C) $\left(\frac{1}{5}, 0\right)$ (D) $\left(\frac{1}{5}, -1\right)$

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67. The distance of the point (1, 1, 1) from the plane passing through the point (1, –1, –1) having normal perpendicular to the lines $\frac{x-4}{1}=\frac{y-3}{-2}=\frac{z-4}{3}$ and $\frac{x-2}{2}=\frac{y+1}{-1}=\frac{z+7}{-1}$ is

(A) $\frac{10}{\sqrt{74}}$ (B) $\frac{20}{\sqrt{74}}$

(C) $\frac{20}{\sqrt{83}}$ (D) $\frac{5}{\sqrt{83}}$

68. If $5(\tan^2x - \cos^2x) = 2\cos2x + 9$, then the value of $\cos4x$ =

(A) $-\frac{7}{9}$ (B) $-\frac{3}{5}$

(C) $\frac{1}{3}$ (D) $\frac{2}{9}$

69. If ω is a non real cube root of unity, then $1(2-w)(2-w^2) + 2(3-w)(3-w^2) + 3(4-w)(4-w^2) + \ldots\ldots + (n-1)(n-w)(n-w^2)$ =

(A) $\left(\frac{n(n+1)}{2}\right)^2 - n$ (B) $\left(\frac{n(n-1)}{2}\right)^2 - n$

(C) $\left(\frac{n(n+1)}{2}\right) - n$ (D) $\left(\frac{n(n-1)}{2}\right)^2 - 1$

70. Let C_1 and C_2 are two circles whose equations are $x^2 + y^2 = r^2$ and $(x-r)^2 + y^2 = r^2$ respectively. If AB is their common chord and a straight line through A meets the circle C_1 at D and circle C_2 at E, then the $\angle DBE$ is

(A) 60° (B) 90°

(C) 30° (D) 120°

71. Let ABC be a triangle with AB = AC. D is the mid point of BC. E is the foot of perpendicular from D to AC and F is mid point of DE, then AF is

(A) perpendicular to BE (B) parallel to BE

(C) perpendicular to DE (D) none of these

72. The range of the value of a for which the line y + x = 0 bisects the two distinct chords drawn from the point $\left(\frac{1+\sqrt{2}a}{2}, \frac{1-\sqrt{2}a}{2}\right)$ to the circle $2x^2 + 2y^2 - \left(1+\sqrt{2}a\right)x - \left(1-\sqrt{2}a\right)y = 0$

(A) (–2, 2) (B) $(-\infty, -2) \cup (2, \infty)$

(C) $\left(-\sqrt{2}, \sqrt{2}\right)$ (D) $\left(-\infty, -\sqrt{2}\right) \cup \left(\sqrt{2}, \infty\right)$

73. Let $\vec{A} = 2\hat{i} + k, \vec{B} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{C} = 4\hat{i} - 3\hat{j} + 7\hat{k}$, the vector $\vec{R}$ is satisfies the conditions $\vec{R} \times \vec{B} = \vec{C} \times \vec{B}$ and $\vec{R} \cdot \vec{A} = 0$, then the vector $\vec{R}$ is

(A) $-\hat{i} - 8\hat{j} + 2\hat{k}$ (B) $2\hat{i} - 8\hat{j} + 4\hat{k}$

(C) $\hat{i} - 4\hat{j} - 6\hat{k}$ (D) none of these

74. The sum of the series $1^3.2 + 2^3.3 + 3^3.4 + \ldots\ldots n^3(n+1)$ for n = 9 is

(A) (330) (298) (B) (296) (220)

(C) (760) (810) (D) (66) (263)

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75. Range of the function $f(x)=\frac{\sin\left(\frac{x}{2}\right)}{\sqrt{1+\tan^2\left(\frac{x}{2}\right)}}-\frac{\cos\left(\frac{x}{2}\right)}{\sqrt{1+\cot^2\left(\frac{x}{2}\right)}}$ is $x \neq \pi$, $x \in (0, 2\pi)$

(A) {0} (B) [–1, 1]
(C) [0, 1] (D) [–1, 0]

76. If $f(x) = x^5 + 5x - 1$, then $\int_5^{41} \frac{dx}{\left[f^{-1}(x)\right]^5 + 5f^{-1}(x)} =$

(A) 0 (B) ln3
(C) ln4 (D) ln7

77. The value of $\int_0^1 \frac{\sqrt{x}dx}{(1+x)(1+3x)(3+x)}$ is

(A) $\frac{\pi}{8}\left(1-\frac{\sqrt{3}}{2}\right)$ (B) $\frac{\pi}{4}\left(1-\frac{\sqrt{3}}{6}\right)$
(C) $\frac{\pi}{8}\left(1-\frac{\sqrt{3}}{6}\right)$ (D) $\frac{\pi}{4}\left(1-\frac{\sqrt{3}}{2}\right)$

78. The differentiable function f satisfies the relation $f(xy) = xf(y) + yf(x)$ for every $x, y \in R^+$ if $f'(1) = 1$, then the function f is
(A) $y = x\ln|x| + 3$ (B) $y = x\log_e|x|$
(C) $2y = x\log_e x$ (D) $2y = x\ln x - 3$

79. A circle is touching two points of the parabola $y^2 = 5 - x$ and the straight line $x = 1$, then the radius of the circle is (given that the circle lies right of y-axis)
(A) 5 (B) 5/2
(C) 3 (D) 3/2

80. If t_1, t_2, t_3 are three variable parameters satisfying the relations $at_2^2t_3^2 = 4b(t_2 + t_3)$ and $at_2^2t_1^2 = 4b(t_1 + t_3)$, then the locus of the point of intersection of tangents to the parabola $y^2 = 4ax$ at $\left(at_1^2, 2at_1\right)$ and $\left(at_3^2, 2at_3\right)$ is
(A) $y^2 = 4bx$ (B) $x^2 = 4by$
(C) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (D) $\frac{y^2}{a^2}+\frac{x^2}{b^2}=1$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If $0 \leq x < \frac{\pi}{2}$, then the minimum value of $\frac{2\sin x + \tan x + 3\cos x}{x + \cos x}$ is

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82. An ellipse with eccentricity $\frac{1}{2}$ passes through the point P(3, 4) whose nearer focus is S(0, 0) the equation of tangent at P on the ellipse is $3x + 4y - 25 = 0$. If a chord through S parallel to the tangent at P intersects the ellipse at A and B, then the length of AB =

83. The coefficient of x^{13} in the expansion of $(1 + x^2)^6 (1 + x^3)^9 (1 + x^4)^{12}$ is

84. Number of integral terms in the expansion of $\left(\sqrt{2} + \sqrt[8]{7}\right)^{128}$ is

85. Let P and Q be 3×3 matrices $P \neq Q$, if $P^3 = Q^3$ and $P^2Q = Q^2P$, then $|P^2 + Q^2|$ is equal to

86. If $a = \log_{45}^{75}$, $b = \log_{135}^{375}$, then the value of $ab + 5(a - b)$ =

87. The sum of the ordinates of the foot of normals drawn from the point (3, 3) to the parabola $y^2 = 8x$ is

88. If LCM of two positive integers (a, b) given ($a < b$) is 60, then the number of possible combinations (a, b) is

89. Let $P(x) = a_0 + a_1x + a_2x^2 + ... a_n$ and $|P(x)| \leq |e^{x-1} - 1|$ for all $x \geq 0$, then the maximum value of $|a_1 + 2a_2 + 3a_3 + ... na_n|$ is

90. Let $\vec{a} = \frac{\sqrt{3}\,\hat{i} - \sqrt{5}\hat{j} + 4\hat{k}}{\sqrt{6}}$ and $\vec{b} = \frac{\sqrt{3}\,\hat{i} - \sqrt{5}\hat{j} - 2\hat{k}}{\sqrt{3}}$, then $\left|\left(3\vec{a} - 2\vec{b}\right)\cdot\left(\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a} + 3\vec{b}\right)\right)\right|$ is

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